

PROGRESS STATION, Antarctica

WMO: 895740

Lat: 69.378S

Lon: 76.391E

Elev: 15

StdP: 101.15

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-33.6	-30.0	-37.2	0.1	-31.5	-35.1	0.1	-28.1	20.6	-11.1	17.7	-8.8	0.8	300	1.089

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.4	4.7	0.8	3.8	0.2	3.0	-0.3	1.2	4.0	0.6	3.2	0.0	2.4	2.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
-0.8	3.5	0.7	-1.8	3.3	0.5	-2.7	3.0	0.4	11.7	4.1	10.7	3.3	9.7	2.5	3.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	13.2	11.2	-35.8	5.9	5.5	1.3	-39.8	6.9	-43.0	7.6	-46.1	8.4	-50.1	9.3		
(5)			-35.5	2.0	4.8	0.7	-39.0	2.5	-41.9	2.9	-44.6	3.3	-48.1	3.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.8	0.4	-3.0	-8.7	-12.6	-14.6	-15.2	-16.3	-15.8	-15.1	-11.8	-4.5	-0.2	(6)
(7)		DBStd	7.71	1.59	2.40	4.05	4.54	5.88	5.62	7.08	6.26	5.99	4.36	3.28	2.07	(7)
(8)		HDD10.0	7231	297	363	579	679	762	755	814	799	753	677	434	318	(8)
(9)		HDD18.3	10272	556	597	838	929	1020	1005	1073	1057	1003	936	684	576	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.4	3.1	4.3	4.8	4.4	4.9	4.9	4.5	5.0	4.7	4.4	4.1	3.4	(14)
(15)	Precipitation	PrecAvg	336	38	39	32	21	23	24	27	24	21	22	34	31	(15)
(16)		PrecMax	450	107	72	69	54	53	56	74	80	53	52	85	61	(16)
(17)		PrecMin	170	12	19	11	8	8	6	1	3	1	6	12	9	(17)
(18)		PrecStd	52	19	15	14	11	11	10	15	15	11	10	17	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	4.2	0.5	-1.6	-0.2	-2.1	-1.6	-2.1	-2.5	-1.3	2.6	6.0	(19)
(20)			MCWB	2.2	0.3	-1.8	-3.7	-2.8	-4.0	-3.8	-3.4	-4.4	-4.2	-0.9	1.7	(20)
(21)		2%	DB	4.8	2.1	-1.4	-3.8	-3.3	-4.3	-3.9	-4.8	-4.3	-3.1	1.7	4.9	(21)
(22)			MCWB	0.9	-1.3	-3.4	-5.2	-4.8	-5.8	-5.6	-6.5	-5.9	-5.4	-1.4	0.9	(22)
(23)		5%	DB	3.9	1.2	-2.4	-5.0	-4.8	-6.2	-5.5	-6.4	-5.7	-4.3	0.8	4.0	(23)
(24)			MCWB	0.3	-1.6	-4.2	-6.3	-6.2	-7.6	-6.7	-7.9	-7.3	-6.1	-2.0	0.2	(24)
(25)		10%	DB	3.2	0.4	-3.4	-6.3	-6.4	-7.8	-7.2	-7.8	-7.1	-5.5	0.0	3.1	(25)
(26)			MCWB	-0.1	-2.1	-5.0	-7.8	-7.7	-9.1	-8.3	-9.0	-8.7	-7.2	-2.5	-0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	0.8	-1.3	-3.2	-2.7	-3.8	-3.6	-3.4	-4.1	-3.7	-0.1	2.1	(27)
(28)			MCDWB	5.8	2.6	0.4	-2.0	-1.0	-2.2	-2.3	-2.5	-3.0	-2.3	1.7	5.4	(28)
(29)		2%	WB	1.3	-0.3	-3.0	-4.9	-4.6	-5.6	-5.1	-6.3	-5.8	-5.0	-0.9	1.2	(29)
(30)			MCDWB	4.0	1.1	-1.9	-3.8	-3.3	-4.5	-4.1	-5.0	-4.3	-3.4	0.8	4.3	(30)
(31)		5%	WB	0.7	-1.0	-3.9	-6.2	-6.1	-7.5	-6.7	-7.7	-7.1	-6.1	-1.5	0.5	(31)
(32)			MCDWB	3.3	0.7	-2.7	-5.0	-4.9	-6.3	-5.6	-6.6	-5.9	-4.4	0.4	3.4	(32)
(33)		10%	WB	0.2	-1.9	-4.9	-7.8	-7.6	-8.9	-8.3	-8.9	-8.6	-7.1	-2.3	-0.1	(33)
(34)			MCDWB	2.6	0.2	-3.4	-6.5	-6.4	-7.7	-7.1	-7.7	-7.2	-5.5	-0.2	2.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.4	3.2	3.3	3.9	4.5	4.5	5.0	5.2	4.5	4.3	4.0	3.6	(35)
(36)			MCDBR	3.9	3.7	3.1	4.0	4.0	5.2	4.5	4.7	4.4	4.6	4.2	4.3	(36)
(37)			MCWBR	3.0	3.2	2.8	3.9	3.7	4.9	4.2	4.6	4.0	4.2	3.6	3.2	(37)
(38)		5% WB	MCDBR	3.6	3.0	3.2	4.1	4.0	5.5	4.4	5.1	4.8	4.5	3.8	3.9	(38)
(39)			MCWBR	3.0	3.1	3.3	4.1	3.9	5.3	4.2	5.1	4.5	4.3	3.4	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.246	0.219	0.162	0.121	N/A	0.124	0.158	0.202	0.217	0.242	0.258	(40)	
(41)		taud	2.314	2.344	2.305	2.103	1.903	N/A	1.884	2.060	2.221	2.235	2.245	2.262	(41)	
(42)		Ebn at Noon	992	943	858	650	90	0	59	636	863	971	1004	1006	(42)	
(43)		Edh at Noon	106	88	66	34	3	0	3	35	69	96	113	117	(43)	
(44)	All-Sky Solar Radiation	RadAvg	8.05	5.23	2.56	0.75	0.06	0.00	0.01	0.38	1.87	4.51	7.42	9.08	(44)	
(45)		RadStd	0.29	0.25	0.12	0.04	0.00	0.00	0.00	0.02	0.10	0.17	0.25	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page